



PROJECT PROCESS DOCUMENT

Secure Website to Display Google Drive Images on an
Interactive Map

Developed By: DevZoro

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1. Introduction

This Process Document explains the complete workflow followed throughout the development of the project “Secure Website to Display Google Drive Images on an Interactive Map with Filters.” The goal was to build a fully automated, secure, and user-friendly system that retrieves images from Google Drive, extracts their GPS metadata, and displays them interactively on a map.

2. Requirement Analysis

Core Requirements Identified:

- Secure login system for authorized users
- Automatic syncing of images from Google Drive
- Extract GPS metadata (latitude, longitude, timestamp)
- Display images on a map with smart clustering
- Year-based filter for better navigation
- Responsive UI across iOS, Android, Windows, and macOS
- Deployment on cloud infrastructure
- Admin-level features such as resyncing and user control

3. Planning & Architecture

Project Architecture:

Frontend (ReactJS + Google Maps API)

Backend (Node.js + MongoDB)

Cloud Deployment (Vercel + Google API)

System Flow:

1. User logs in
2. Backend fetches images from Google Drive
3. Metadata is extracted
4. Clean data stored into MongoDB
5. Frontend loads data on Google Maps
6. Filter options generated dynamically

4. Development Process

Step 1 — Authentication System

Step 2 — Google Drive Integration

Step 3 — Metadata Extraction

Step 4 — Map Interface Development

Step 5 — Filtering System

Step 6 — Responsive UI/UX

Step 7 — Additional Enhancements

Step 8 — Deployment

5. Testing & Quality Assurance

Device Testing:

- iPhone
- Android
- MacBook
- Windows

Browser Testing:

- Chrome
- Edge
- Firefox
- Safari

Load Testing:

Tested with 5,000+ images

6. Final Deliverables

- Complete source code
- Deployment documentation
- Admin & user guide
- MongoDB structure
- Google API instructions
- Working production build
- Cloud-based deployment
- Training on adding users and syncing photos

7. Conclusion

This project was successfully completed by DevZoro, delivering a robust, scalable, and user-friendly platform.